Proceedings of the 2017 IEEE 21st International Conference on Computer Supported Cooperative Work in Design

A New Subtle Hand Gestures Recognition Algorithm Based on EMG and FSR

Bo Wan
School of Computer Science
Xidian Universit
Xi'an, China
wanbo@xidian.edu.cn

Rongrong Wu
School of Computer Science
Xidian University
Xi'an, China
aawurongrong@163.com

Kaihuo Zhang
School of Computer Science
Xidian University
Xi'an, China
zhangkaihuo@gmail.com

Lanxin Liu
School of Computer Science
Xidian University
Xi'an, China
18710915593@163.com

***Abstract*—The paper proposes a new algorithm that combines the EMG (Electromyography) signals and pressure signals for detecting subtle hand gestures. First, the algorithm uses Myo armband to acquire the forearm EMG signals and the algorithm uses an array of five force sensitive resistors (FSRs) to acquire the muscle expansion pressure signals of the back of the hand. Then the algorithm has an activity detection on the collected signals. The algorithm has selected the top 50 best features of a gesture via the open source tool scikit-learn. After that the cross-validation is used to select the best parameter k of kNN(k-Nearest Neighbor) algorithm. Finally, the selected k of kNN algorithm is used for gesture classification in real time. The experiment shows that the algorithm presented in this paper has a classification accuracy of 96.05% for 21 pre-defined hand gestures, and the combination of EMG and FSR is significantly more accurate for subtle gesture recognition than the one using EMG only and using FSR only.**

Keywords—subtle gestures; EMG signals; pressure signals

I. INTRODUCTION

With the development of electronic technology and computer technology, traditional input methods such as mouse and keyboard have failed to meet people's requirements. Recently, gesture input has been an important means of human-computer interaction due to the natural and intuitive of gestures. Human-computer interaction based on gestures emphasized more on the user-centered, and it is more suitable for users' habits, which provides a natural and effective human-computer interaction. Since the instantaneity and accuracy of gesture recognition is crucial for natural human-computer interaction, the research of subtle gestures recognition has great significance for natural human-computer interaction. Currently the typical gesture recognition methods can be divided into three categories: the method based on computer vision, the method based on EMG signals and the method based on pressure signals.

A. Gesture Recognition Based on Computer Vision

In gesture recognition based on computer vision, the whole image is generally captured by cameras, and the hand image is segmented from the whole image, and then a series of analysis are made on the hand image. Kumar [1] et al converted the RGB hand image into gray scale image, and then they counted the number of pixels whose gray-level is larger than the given threshold. Finally they had a gesture judgment based on the number of pixels, and they had achieved the recognition of six gestures such as fist, the little finger extension and the flexion of thumb. The algorithm only identified simple hand gestures. Toby [2] et al presented a real-time hand tracking system via a single depth camera. They used a vector to denote the hand information. The vector includes the global scale, orientation and the rotation at each joint, and then they made a match between the input vector and the vector in database. The vector with highest match degree is the class of input gesture. However, training the model takes several hours. The gestures recognition based on computer vision is difficult to distinguish subtle gestures due to the occlusion between fingers.

B. Gesture Recognition Based on EMG Signals

The EMG signals refer to biological signals produced by neuromuscular system when users perform a gesture [3]. Researchers analyze the EMG signals collected from the corresponding muscle groups to achieve the gesture recognition. Edith [4] et al used two EMG sensors placed at the wrist and the forearm. Due to the limitation of EMG amount, they had only realized the recognition of the opening and closing of fingers, and other movements like fast and slow upward action of the forearm. Then the recognition result is used for the control of PPT. Yu [5] et al collected EMG signals which generated by four muscles of forearm, and they had reached the accuracy of 94% of eight kinds of gestures via BP neural network, while the gestures mainly involved the movement of the whole hand but not involved some minor fine finger gestures. Guo [6] et al putted EMG on the human forearm brachioradialis muscle, flexor carpi, extensor flexor and superficial flexor muscle, and they have reached the recognition of 97.53% of six gestures. In general, gesture recognition based on EMG signals requires the researcher to master some physiological knowledge, and the sweat of long time wearing would lead to low sensitivity of sensors.

C. Gesture Recognition Based on Pressure Signals

Pressure signals are the surface pressure generated by muscle expansion. Researchers collect the pressures signals via pressure sensors, and convert the pressure signals to electrical signals. They completed the gesture recognition by analyzing the electrical signals. WristFlex [7] which used an array of force sensitive resistors worn around the wrist, and it's accuracy of 5 kinds of finger pinch gestures is over 80%. WristFlex has low power consumption. Because the wrist area was not fully covered, the misclassification occurred most often in the index finger pinch. And it involved small amount of gestures. Sun [8] used FSRs on the forearm to detect the contraction of muscles, and the results of gesture recognition are used for the control of prosthetic hand. The FSRs had covered almost the whole forearm, which is inconvenient. Li [9] collected the pressure signals on the forearm and classified the signals via SVM, and the classified result is applied to robotic hand controlling. The research had reached the accuracy of 95% among 10 kinds of hand gestures.

McIntosh etc have proposed EMPress[10], a practical hand gesture classification algorithm combining EMG and FSR. EMPress sensed EMG and pressure data at the wrist, and trained the data with SVM. The author grouped 15 gestures into three classes: finger gestures, wrist gestures and other gestures. And they have proved that the EMG is especially suited to sense finger movements, and the pressure is suited to sense the wrist and forearm rotations, and their combination is significantly accurate than using EMG and FSR separately. EMPress has achieved the classification accuracy of 95.8% for 15 kinds of gestures. However, the EMPress have not attempted to classify multiple finger gestures simultaneously, and it's easy to misclassify the adjacent fingers. This is partly due to low electric potential generated by few muscle cells on the wrist, and there are insufficient number of sensors to sense the muscles, and they collected the gesture signals in a low collection frequency.

As a review of previous work of gesture recognition, there are mainly two problems:1. Previous work mainly focused on rough gestures while they have little research on the subtle hand gestures, and it involves few kinds of subtle gestures. 2. The recognition methods based on single signal are limited to itself which have resulted in few recognizable gestures.

To solve the above problems, the paper proposes an algorithm of subtle finger gestures recognition based on EMG and FSR. The paper defines subtle gesture as subtle gestures involving minor hand movements, and it not only includes the movement of the hand, but also the movement of the single finger and the combination of fingers, and the users are not required to perform subtle gestures with too much strength and larger motion. The relatively high misclassification of adjacent fingers of EMPress led us to explore the possibility of putting the FSRs on the back of the hand. The algorithm combines the EMG on the forearm and the FSRs on the back of the hand to identify the classification accuracy of hand gestures including hand movements, single finger movements and multiple finger movements. First, a data glove is made to fix the FSRs. The FSRs is used for the collection of surface pressure distribution on the back of the hand. A cheap wearable device Myo armband is chosen to complete the collection of EMG signals on the forearm. The collected signals are processed by a computer, and the kNN algorithm is used for classification, and then the classification of subtle gestures is completed.

The mainly contributions of the algorithm are as follows:

1. The paper presents a new algorithm for subtle gestures detection, which combines EMG and FSR. The algorithm has achieved the recognition of 21 pre-defined gestures, and its accuracy is over 96%, and it is significantly more accurate for subtle gestures than using EMG only and using FSR only.

2. The algorithm fixed the FSR on the glove, and it works with the Myo armband that forms a new wearable device, which makes the human-computer interaction more natural and it overcomes the limitation of the environment.

3. The paper explores the application scenery, which shows the superiority of the combination of EMG and FSR.

II. System Design

As is shown in Figure 1, the algorithm has intergraded the FSR to a wearable device. The FSR glove and Myo armband work together to complete the identification of subtle gestures.

Myo armband is used for collecting EMG signals on the forearm. The Myo armband is a terminal control equipment developed by Canadian Thalmic Labs Company. Myopoint[11] used Myo armband for cursor control and mouse click, and it had demonstrated that the consumer-level Myo armband is practical for distant pointing and clicking on large displays. The Myo armband has eight EMG sensors and one Inertial Measurement Units (IMU). The Myo official has opened the original EMG data, and the data is sampling at 200HZ. Then the data is sent to the computer via Bluetooth 4.0.

FSR is chose to sense the pressure of the back of the hand. In comprehensive consideration, the algorithm chooses five FSR402 because it is small and convenient to use. It is a thick film polymer sensor, and the resistance value would be decreased while the pressure on the surface increased. And the sensitive characteristic is specially optimized for human touching, so it is suitable for collecting tactile signals. The placement of five FSRs is shown in Figure 2.

Fig. 1. Image of the prototype

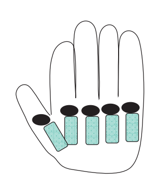

Fig. 2. The placement of FSRs

Five FSRs are fixed on the glove respectively, which has a full contact with the extensor muscle of fingers. Arduino Uno board is chosen to collect the FSRs data. In order to keep pace with EMG signals, the sample rate is set to 200HZ. As is shown in Figure 3, the sample data of Myo armband is transmitted to the computer via Bluetooth. Five FSRs are connected to a circuit board, and the output of the board is connected to the ADC of Arduino Uno. The Arduino Uno board is programmed to read analogue FSR data in sequence, and then sent it to the computer via serial communication for the next processing.

III. Algorithm

Eight channels EMG data and five channels FSR data are recorded in the form of 1D array separately. The algorithm has an activity detection, feature extraction and feature selection on the collected data. Then the k-Nearest Neighbor algorithm is used to classify the gesture vector that we recorded.

A. Active Detection

The active detection is to determine the start point and end point of signals in order to extract the effective gesture signals. The algorithm needs to segment the start point and end point from EMG signals and pressure signals simultaneously. The acquisition experiment shows that the amplitude of EMG signals between two gestures decreased significantly, so the algorithm segment the pressure signals active detection based on the EMG active segmentation detection. Currently EMG activity segment mainly includes moving average method [12], artificial neutral network [13] and short-time Fourier [14]. In the consideration of real-time system, the moving average method is chose for EMG signals activity segment.

To have a clear observation, the Figure 4 shows the first four channels of EMG signals and the first three channels of pressure signals. As is shown in Figure 4, the corresponding channels of EMG signals and pressure signals have different changes when users make different gestures, providing a perfect basis for the feature extraction in the next step.

B. Feature Extraction and Feature Selection

The purpose of feature selection is to find a set of data that can represent the feature of the gesture signals, and the distance between different gesture class is as small as possible which has a better clustering, and the inter-distance of gesture class is as large as possible which has a better separability.

This paper has extracted the time domain, frequency domain and time-frequency domain features of the gesture. This paper has extracted RMS (root mean square) , SD (standard deviation) , The fourth order AR model coefficients, MPF (mean power frequency) and MF (median frequency) from EMG signals and pressure signals. There are eight channels of EMG signals and five channels of pressure signals in our system, so the algorithm has extracted 104 features from the original signals.

The scikit-learn is used for feature selection and the classification of the next step. Scikit-learn is a Python module integrating a wide range of machine learning algorithms, and it emphasizes on the easiness of use and performance. The training data are scaled between -1 and 1 to hasten the optimal solution of gradient descent, and it is possible to improve the accuracy of the result. And the testing data are similarly processed for prediction. The 50th highest scores of extracted features are selected for the next classification. So a gesture can be represented as a 1*50 dimensional feature vector.

C. Gesture Classification with kNN

The original kNN algorithm was proposed by Cover and Hart in 1968, and then it was deeply analyzed and studied theoretically, which is one of the most important nonparametric methods [15]. The reason why the algorithm chooses kNN is because the kNN is easy to implement, and our training samples are small, so it will not cause too much computation.

The performance of kNN algorithm mainly depends on the selection of a “good value” of k. If k is too small, the algorithm will get few neighbors, and it will reduce the classification accuracy and cause over fitting. However, if k is too large, the computation will be large. This paper obtains the optimal k via cross validation. Through 10-folds cross validation on our data sets, the algorithm calculates cross validation accuracy of the different values of the parameter k. Then the algorithm selects k with the highest cross validation accuracy.

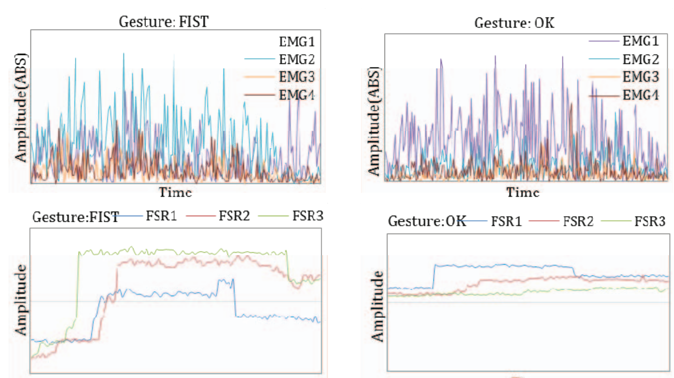

Fig. 4. The EMG signal (top) and the pressure signal (bottom) of two kinds of gestures

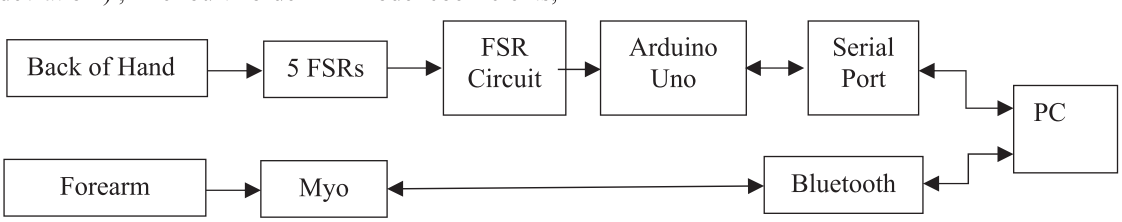

Fig. 3. Data collection system

For example, the algorithm has tested the k vary from 1 to 15, the cross validation accuracy is shown in Figure 5. The algorithm selects k = 8 which has the highest cross validation accuracy as the optimal parameters. The kNN algorithm does not need to generate the model, and it simply stores the original training data. When a new gesture vector is coming, the kNN will calculate the distance of 8 nearest neighbors of training data to achieve the gesture classification.

IV. EXPERIMENT

The paper has three experiment conditions: using EMG only, using FSR only and using EMG and FSR together. The paper recruited 30 participants, 12 female and 18 male, who have no any muscle disease to complete the experiment and their average age is 30 years(SD: ±5.4). They are all right-handed. The participants are divided into three groups randomly. The participants should complete 21 pre-defined gestures. The definition of the gestures is shown in Figure 6.

A. Experiment Settings

The experimental environment is HP desktop 2.35GHZ, Xeon Intel, 4GB memory.

In order to have a contrast with the results by using EMG only, using FSR only and using EMG and FSR together, the experiment is divided into three experiments. Ten participants in group 1 wear Myo armband only in Experiment 1, and ten participants in group 2 wear FSR glove only in Experiment 2, while the other ten participants in group 3 wear Myo armband and FSR glove in Experiment 3. The experimental process in Experiment 1, Experiment 2 and Experiment 3 is the same. The Myo armband and FSR glove are in the right hand.

The paper has an Exercise Task followed by a Formal Task. The aim of the Exercise Task is to make the participants being familiar with the experimental procedures and the gestures needed to perform. The Exercise Task has two blocks. Each block has to perform 21 pre-defined gestures. The sequence of the gestures in each block is random, and it has an interval of at least two seconds between two gestures, and it has a 3-minute rest after one block is completed. The experiment provides a paper that records the sequence of gestures due to the large amount of gestures. The Formal Task will start after the Exercise Task. The Formal Task consists of twenty blocks of training phase and ten blocks of testing phase. When the participants perform the gestures in training phase, the screen will not show the classified result. However, the screen will show the recognition result after the participants perform the gesture in testing phase. The rest of Formal Exercise is the same as in the Exercise Task.

In summary, for each participant, there are 420 gestures for training and 210 gestures for testing.

B. Experimental Results and Analysis

The average recognition accuracy in Experiment 1 is 81.71%(SD: ±4.0%), the Experiment 2 is 79.52%(SD: ±5.31%) and the Experiment 3 is 96.05%(SD: ±4.12%).

In order to determine whether the difference is statistically significant between Experiment 1, 2, 3, the paper have a t-test on the recognition accuracy. At the significant level $\alpha = 0.05$, the difference between Experiment 1 and Experiment 2 is not significant ($P(T<=t) = 0.192$), and the difference between Experiment 1 and Experiment 3 is significant(($P(T<=t) =0.000016$) and the difference between Experiment 2 and Experiment 3 is significant(($P(T<=t) =0.000027$). The t-test shows that the recognition accuracy of Experiment 3 is significantly improved compared with Experiment 1 and Experiment 2 in the condition of the significant level $\alpha = 0.05$.

The confusion matrix in Figure 7 shows the performance of using EMG only for 21 pre-defined gestures. Using EMG only has an ideal results on the whole movement of the hand such as UP, DOWN, LEFT, RIGHT, FIST and SPREAD.

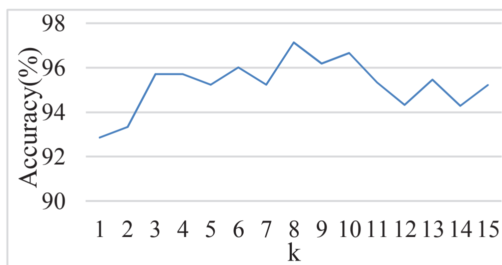

Fig. 5. The 10-fold cross validation accuracy of different k

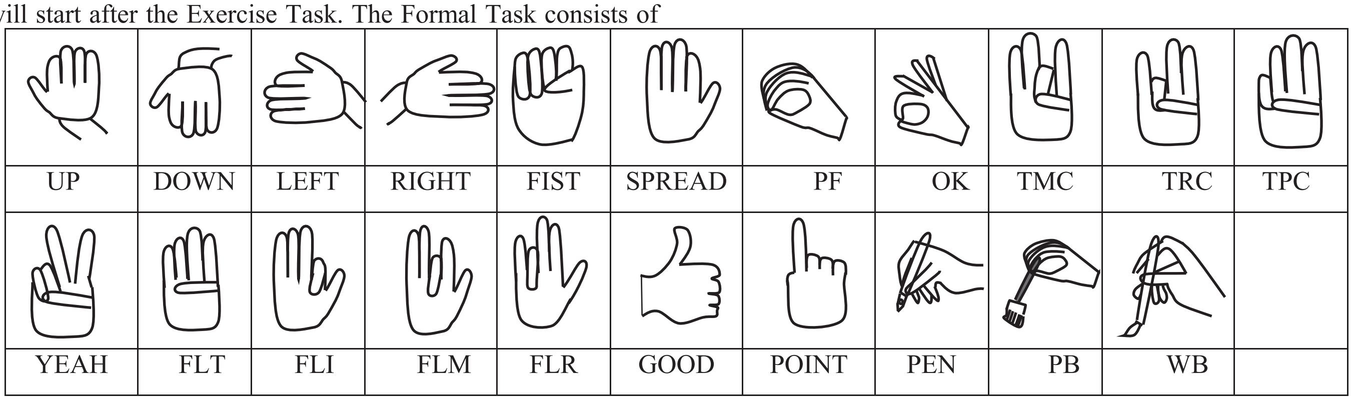

Fig. 6. 21 Pre-defined gestures

However, in Figure 7, it is not ideal for subtle movements in fingers, and the misrecognition mainly focuses on the movement of a single finger and the combination of fingers. The whole hand movement recognition rate varies from 90% to 95%, while the gestures involved fingers varies from 68% to 83%. It has the higher number of misrecognition on the thumb and middle in circle (TMC), flexing the middle (FLM), and writing brush-holding gesture (WB). The misclassification mainly occurres on the adjacent finger gestures. For example, the kNN misclassified 9 thumb and middle in circle (TRC) gestures as TMC gestures, and the FLR gestures have the highest number of misclassification to FLM. This is because the EMG signals caused by subtle gesture are slight. The writing brush-holding gesture has the lowest recognition accuracy, the possible reason is because the writing brush-holding gesture is relatively complex, and it needs the five fingers to work together.

Figure 8 shows the confusion matrix of using FSR only. Compared to the whole hand movements, the recognition of the gestures involved finger is higher. Compared to use EMG only, using FSR only has a relatively lower recognition on the whole hand movements. This is mainly because the whole hand movements involve the rotation of the palm and with little finger move, so it causes the recognition of the whole hand movement not satisfied when using FSR only. The recognition of the majority of gestures involved fingers is higher than using EMG only. However, it is not significant. And the recognition of some finger gestures has decreased. For example, the recognition of the thumb and pinky in circle (TPC) and painting brush-holding gesture (PB) is slightly decreased. It perhaps because FSRs moves while the participant perform the gesture.

As can be seen in Figure 9, the combination of EMG and FSR is significantly more accurate for gestures than using EMG only and using FSR only, the combination has a positive effects on the movement of the whole hand, a single finger and more than one finger. The combination of EMG and FSR has reached 100% accuracy in FIST gesture. The writing brush-holding gesture has the lowest classification accuracy of 89%. Because the ordinary people rarely use a writing brush, so the writing brush-holding gestures is not standard which caused the low classification accuracy. The misclassification of adjacent fingers is greatly reduced in Figure 9, however, the kNN still misclassified FLM to flexing the index (FLI) 3 times. This is mainly because other fingers would be flexed while flex the middle, so it led to misclassification. And there is no obvious misclassification for any particular gestures.

Having compared these three confusion matrix, the combination of EMG and FSR has great improvements on the gesture recognition. For example, gesture TMC, TPC, FLR and WB have the lower recognition accuracy in Figure 7 and Figure 8. However, these gestures have a satisfying recognition accuracy in Figure 9.

Table 1 shows the comparison of the algorithm the paper proposed and other algorithms. As can be seen in Table 1, the algorithm proposed in this paper has superior performance compare to other algorithms. Our experiment shows that the combination of EMG and FSR has significantly improved the

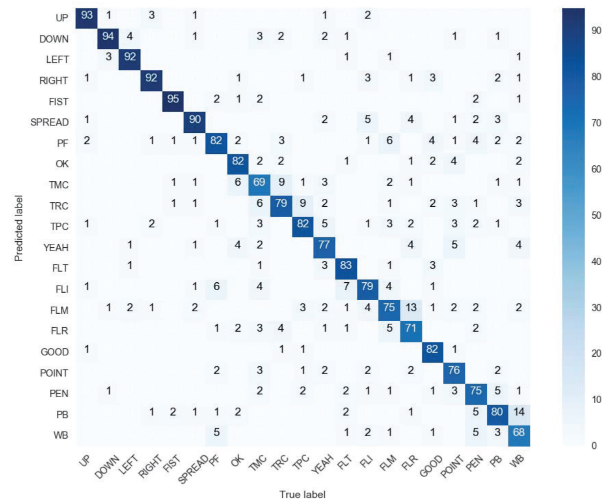

Fig. 7. Confusion matrix using EMG only

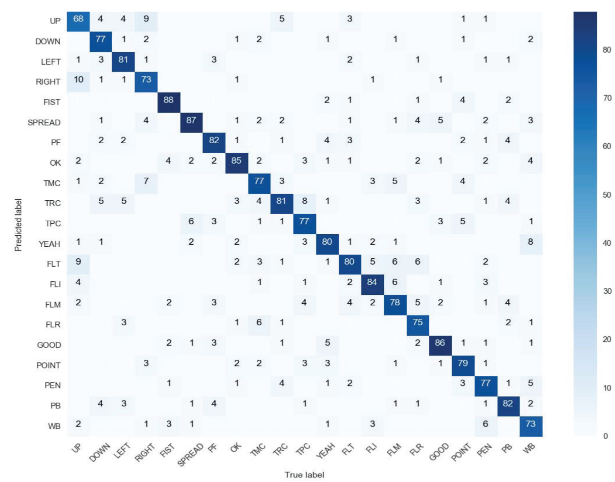

Fig. 8. Confusion matrix using FSR only

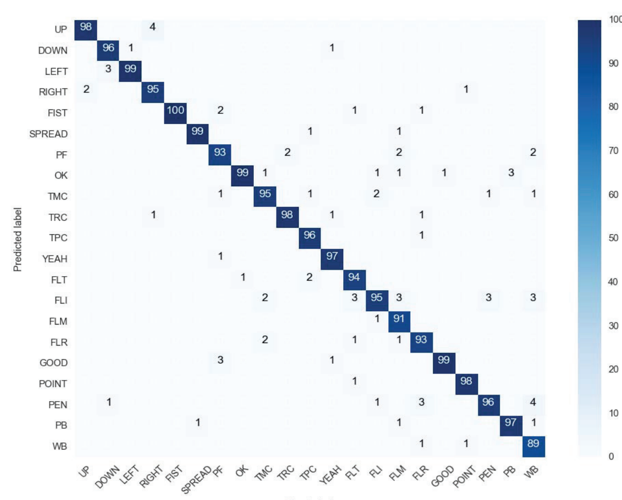

Fig. 9. Confusion matrix using EMG and FSR

TABLE I. THE COMPARISON TO OTHER ALGORITHMS

Algorithm	Real-time Accuracy	The Number of Gestures	The Type of Gestures
Empress	95.8%	15	Wrist gestures, finger gestures and others
WritsFlex	80.5%	6	Pinch gestures
YU's Algorithm	>94%	8	Hand gestures and finger gestures
Our Algoritm	96.05%	21	Hand gestures and finger gestures

accuracy of subtle gestures, and the standard deviation has been reduced compared to use FSR only. It has a good feasibility and effectiveness in subtle gestures.

V. TYPICAL APPLICATION SCENARIOS

Gesture recognition can be used to detect the pen-holding gesture of the users, and save the writing content as electronic documents with the use of Wii Remote. The system recognizes the pen-holding gesture and tracks the pen with Wii Remote. Different pen-holding gestures caused different writing effects, and the system saves it in the form of electronic document.

While the user is writing, the system would recognize the pen-holding gestures of the user. If the pen-holding gesture is identified as a pen, the system would render the write effect as the pen people usually use. If the pen-holding gesture is identified as a writing brush, then the writing effect is a brush pen, and the system would render the thickness and color depending on the strength of the users. When the user writes hardly, the write lines would be thick and the color would be dark. Otherwise the write lines would be fine and the color would be shallow. If the pen-holding gesture is identified as a painting brush, the current writing is rendered as a painting brush. The system adjusts the color depth of the painting. The user switches the color of painting with the gestures left and right. When the user performs the gesture OK, the system would save the writing automatically.

VI. LIMITATIONS AND FUTURE WORK

Myo armband couldn't be integrated with FSRs glove, which has caused the inconvenience. The algorithm plans to explore the possibility of putting the EMG sensors on the wrist and placing the FSRs on the back of the hand. The EMG sensors collect the EMG signals on the wrist, and fix the EMG sensors and FSRs on the glove to form a wearable device. Then the algorithm can complete the gesture recognition with a data glove.

Finally, the algorithm has not considered the motion of the forearm, so the paper considers using acceleration for detecting more gestures.

VII. CONCLUSION

The paper has proposed a new wearable device, which can be used in real time and natural human-computer interaction. The algorithm this paper proposed has combined EMG and FSR, and used kNN algorithm to complete the real time recognition of subtle gestures. Then the paper has an experiment on 21 pre-defined subtle hand gestures. The experiment shows that the combination of EMG and FSR has reached the accuracy of 96.05% for subtle gestures, which proved the feasibility of the combination of EMG and FSR. The paper also describe the typical application of the gesture recognition based on EMG and FSR. The application shows that our algorithm has a great potential on the new wearable device of gesture recognition.

ACKNOWLEDGMENT

This work is supported by "The Fundamental Research Funds for the Central Universities" (Grant No. JB160305).

REFERENCES

[1] P.Vijaya Kumar , N.R.V.Praneeth, Sudheer.V. Hand and finger gesture recognition system for robotic application. In Proceedings of the International Journal of Computer Communication and Information System, 2010.

[2] Sharp Toby, et al. Accurate, robust, and flexible real-time hand tracking. ACM Conference on Human Factors in Computing Systems, 2015, pp. 3633-3642.

[3] Wang J, Jin X G. sEMG signal analysis method and its application research [J]. China Sportence & Technology, 2000.

[4] RM Edith, AB Haripriya. Gesture recognition using real time EMG. In Proceedings of the Innovations in Information, Embedded and Communication Systems, 2015, pp.1-3.

[5] Qing YU, Yang J, Chen X. Pattern Recognition of Gesture's Surface Myoelectrogram signal based on BP Neural Networks [J]. Journal of Biomedical Engineering Research, 2009.

[6] GUO Yina, WANG Qinghua, ABRAHAM Ajith. Real-time electromyography hand gesture recognition model based on flexible neural trees. Computer Engineering and Applications, 2012, pp. 207-210.

[7] Artem Dementyev and Joseph A. Paradiso. 2014. WristFlex: Low-power gesture input with wrist-worn pressure sensors. In Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology. ACM, New York, NY, USA, pp. 161–166.

[8] Sun C, Jiang L, Yang D . Recognition system for hand gestures based on FSR sensors [J]. Machinery & Electronics, 2009.

[9] N Li, B Liu, H Huo, Y Ye, L Jiang. Human-Machine Interaction control based on force myograph and electrical stimulation sensory feedback for multi-DOF robotic hand [J]. ROBOT, 2015, pp, 718-724.

[10] Mcintosh, Jess, et al. EMPress: Practical handgesture classification with wrist-mounted EMG and pressure sensing." CHI Conference, 2016, pp. 2332-2342.

[11] Haque, Faizan, M. Nancel, and D. Vogel. Myopoint: Pointing and clicking using forearm mounted electromyography and inertial motion sensors. CHI Conference 2015, pp. 3653-3656.

[12] Chen X, Zhang X,. Multiple hand gesture recognition based on surface EMG signal[C]. The International Conference on Bioinformatics and Biomedical Engineering. 2007, pp. 506-509.

[13] He Lesheng. The research of human-computer interface based on EMG [D]. Southeast University, 2006.

[14] Chen X, Zhang X. The research of pattern recognition based on surface EMG of multi-class gesture. 2007 Academic Annual Conference of the China Federation of Biomedical Engineering. Xi'an, China, 2007, pp.213-216.

[15] Cover T M. Rates of convergence for nearest neighbor procedures[C]. Hawaii International Conference on System Sciences. 1968.